AGRIVISION: Automated Plant Disease Recognition Using Image-Based Deep Learning

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***Abstract*—Plant diseases are a major concern for food security across the globe, resulting in economic losses and making agricultural sustainability a concern. Current approaches for detecting plant diseases are laborious and require a long time, making it imperative for a more efficient approach to emerge. AGRIVISION, a computer model using image based deep learning for automated plant disease detection, is a quick and efficient system for plant disease detection. AGRIVISION uses a lightweight MobileNetV2 model, trained on a PlantVillage dataset with about 54000 images belonging to 38 classes. Based on the results it is proven that the AGRIVISION computer model performed outstandingly on plant disease detection with an accuracy of 98.42% and a heavily weighted precision, recall, and F1-Score of over 98%. It has been deployed on a website for its real world applicability. Hence it plays a significant role in achieving Sustainable Development Goal 2: Zero Hunger with a more accurate approach to detecting plant diseases.**

Keywords*—*plant disease detection, deep learning, computer vision, MobileNetV2, agricultural technology, precision agriculture, sustainable development

I. Introduction

Plant diseases are a significant threat to global food security, causing massive losses in agricultural productivity [1]. As the world population grows and climate change accelerates, food security becomes more fragile, and the need for effective solutions to monitor, detect, and prevent plant/crop diseases is urgent [2]. According to the Food and Agriculture Organization of the United Nations (FAO), up to 40% of global crop production is lost annually due to plant pests and diseases. These crop losses result in economic damages exceeding USD 220 billion globally, with invasive insect species alone accounting for at least USD 70 billion [3].

FAO states that food security consists of four main dimensions: food availability, access, utilization, and stability [4]. Yield reductions in agricultural production resulting from plant pathogens pose a significant threat to these dimensions, which in turn poses a direct threat to food security [5], [6]. Traditional methods of disease management, such as manual inspection, are often time-consuming, prone to error, and inaccessible to small farmers. As a result, many farmers fail to detect infections early, leading to widespread crop damage and lower productivity [7].

To address this challenge, AGRIVISION: Plant Disease Detection System, was developed to support Sustainable Development Goal (SDG) 2: Zero Hunger, which aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture. AGRIVISION leverages Computer Vision to create a model capable of detecting plant diseases through automated image analysis. By enabling early and accurate identification of plant diseases, the system empowers farmers to take immediate action to protect their crops, reduce losses, and promote sustainable agricultural practices. Ultimately, this project contributes to enhancing food security, supporting agricultural sustainability, and fostering economic resilience in farming communities worldwide.

II. Related Works

Plant disease detection has been widely explored in several studies. These studies generally aim to improve accuracy, efficiency, and scalability.

A. Convolutional Neural Networks (CNNs)

Earlier studies utilized convolutional neural networks, which, in theory, perform well on image classification tasks due to their ability to automatically extract spatial and hierarchical features. Shrestha et al. [8] proposed a CNN based model to detect common diseases typically found in the leaves of tomato and potato plants. Based on their report, the CNN based model achieved an accuracy of 88.8% on test data, meaning their model can reliably distinguish between healthy and unhealthy leaves.

B. Residual Networks (ResNet)

Other studies focus on scalability and address the weaknesses of earlier models. Mindhe et al. [9] propose a ResNet based model that was trained on the PlantVillage dataset, which contains 54000 plant leaf images. ResNet based models utilize residual connections in order to overcome the vanishing gradient problem that previous CNN based models faced. The study found out that the ResNet based model are able to perform better compared to other models, specifically on accuracy and robustness.

C. Hybrid Approach (CNN + SVM)

Some studies also approached this problem by combining two different models, these are called hybrid models. Haridasan et al. [10] proposed a hybrid model combining CNN and SVM for the classification of paddy plant diseases. These diseases include leaf blight, false smut, brown leaf spot, rice blast, and sheath rot. Based on their testing, the hybrid model was able to achieve a validation accuracy of 91.45%. The model shows an improvement compared to base CNN models, providing greater accuracy and performance. Although hybrid models show promising results, they tend to be more expensive and heavier to run.

III. Methodology

The proposed AGRIVISION system will utilize a Convolutional Neural Network (CNN) based model to automate the classification of plant diseases from digital images.

A. Dataset Preparation

The dataset used for this project is the PlantVillage dataset, which is obtained from Kaggle. It contains approximately 50,000 RGB images categorized into 38 distinct classes, representing both healthy and diseased leaves from 14 crop species, such as tomatoes, corns, potatoes, grapes, apple, and more. The dataset was captured under controlled lighting conditions with uniform backgrounds, making it suitable for supervised image classification tasks. Our group decided to utilize this dataset due to its wide usage in existing plant disease detection research and it is well-labeled.

B. Image Preprocessing

All images are resized to a consistent size and normalized to a $0 - 1$ pixel intensity range then its channel are also standardized (zero mean variance and unit variance relative to ImageNet stats). In addition, data augmentation techniques, such as random rotation, horizontal flipping, zooming, and brightness adjustment are applied to simulate real-world environmental variations and reduce overfitting problem. The dataset already comes with 2 splits which is training and test data. These preprocessing steps are expected to help the model learn robust features and enhance the performance of the model when exposed to unseen data.

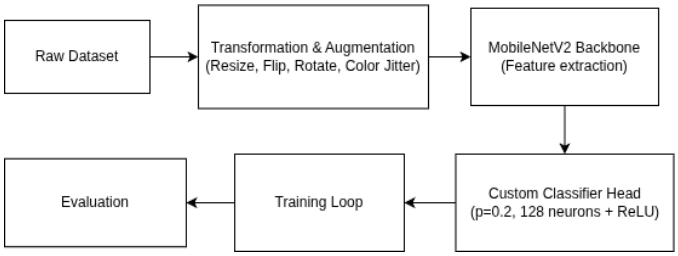


Fig. 1: Overall Training Pipeline of the AGRIVISION

C. Model Architecture and Model Training

As illustrated in Figure 1, the training process of the AGRIVISION system begins with the raw PlantVillage dataset, which undergoes image transformation and data augmentation, including resizing, random flipping, rotation, and color jittering. These steps improve the model's robustness and reduce overfitting by exposing the network to diverse visual variations.

The augmented images are then passed into the MobileNetV2 backbone, which serves as a feature extractor using pre-trained weights from the ImageNet dataset. MobileNetV2 is selected due to its lightweight architecture and favorable trade-off between accuracy and computational efficiency, making it suitable for deployment in low-resource environments such as web servers or mobile devices. In this project, a transfer learning approach is used, where the convolutional base layers of MobileNetV2 are frozen to retain previously learned visual features. The top layers are replaced with custom dense layers consisting of:

- A Global Average Pooling layer to reduce dimensionality.
- A Dense layer (128 neurons, ReLU activation) for feature extraction.
- A Softmax Output layer (38 neurons) corresponding to the disease classes.
- This configuration enables effective learning while minimizing training time and computational cost.

During the training loop, the model is optimized using a suitable optimizer, such as Adam or Stochastic Gradient Descent (SGD), with categorical cross-entropy employed as the loss function. The model parameters are iteratively updated through backpropagation over multiple epochs. Key hyperparameters, including learning rate, batch size, and number of epochs, are determined empirically to achieve optimal classification performance on the PlantVillage dataset.

The detailed parameter update process during training is further illustrated in Figure 2.

D. Inference Pipeline Diagram

As illustrated in Figure 3, the inference pipeline of the AGRIVISION system describes the process of classifying unseen leaf images using the trained model. The pipeline begins with an input image captured by the user, which is first converted into an RGB format and resized to match the input dimensions required by MobileNetV2. The image is then

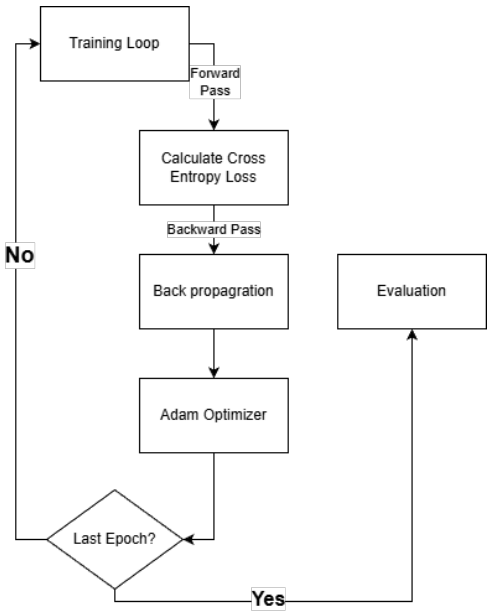


Fig. 2: Training Loop and Optimization Process

normalized to ensure consistency with the data distribution used during model training.

Following preprocessing, the image is transformed into a tensor and reshaped into a batch format before being forwarded through the trained MobileNetV2 network. During inference, the convolutional base and the custom classification head operate in evaluation mode, producing class probability scores through the Softmax layer.

The predicted class is determined by selecting the highest probability output, which corresponds to a specific plant disease or healthy leaf category. This prediction is subsequently mapped to a human-readable label and returned to the user along with its confidence score. This inference pipeline enables efficient and reliable disease identification, making the AGRIVISION system suitable for real-time deployment in practical agricultural applications.

E. Evaluation Metrics

The model performance is evaluated using accuracy, precision, recall, and F1-score, supported by a confusion matrix to analyze class-wise performance.

1) Accuracy

Accuracy measures how often the model correctly identifies the disease class out of all predictions. Accuracy shows the overall proportion of correctly classified leaf images (both healthy and diseased). However, accuracy alone can be misleading if the dataset is imbalanced (e.g., many healthy leaves but few diseased ones).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where:

- TP (True Positives): The number of diseased leaves correctly identified as diseased.
- TN (True Negatives): The number of healthy leaves correctly identified as healthy.
- FP (False Positives): The number of healthy leaves incorrectly classified as diseased.
- FN (False Negatives): The number of diseased leaves incorrectly classified as healthy.

2) Precision

Precision quantifies how many of the leaves predicted as having a certain disease are actually infected with that disease. High precision means the model makes few false alarms (it rarely predicts a disease when the leaf is actually healthy or infected by another disease).

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

3) Recall

Recall measures how well the model detects actual diseased leaves. High recall indicates that the model successfully captures most of the true disease cases, which is important for early detection and minimizing missed infections.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

4) F1-Score

The F1-score provides a balance between precision and recall. It is especially useful when there's an uneven class distribution, ensuring that both the accuracy of positive predictions and completeness of disease detection are considered.

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{4}$$

IV. Implementation and Results

This section will present and analyze the results obtained throughout the training and evaluation process of the model. Results from all metrics demonstrated stable convergence and strong generalization across all boards.

A. System Details

The proposed plant disease detection model is implemented in Python using deep learning frameworks suitable for computer vision tasks. Model training is conducted on a desktop workstation equipped with an NVIDIA RTX 5060 Ti GPU with 16 GB of VRAM, enabling efficient training of convolutional neural networks.

In addition to offline experimentation, the trained model is deployed on a web-based platform hosted on a Virtual Private Server (VPS) to evaluate real-world usability. The VPS is configured with 2 GB of memory and 35 GB of storage, reflecting a resource-constrained deployment environment. This deployment setup demonstrates the practicality of the proposed approach for real-world applications, including web and mobile-based disease diagnosis systems.

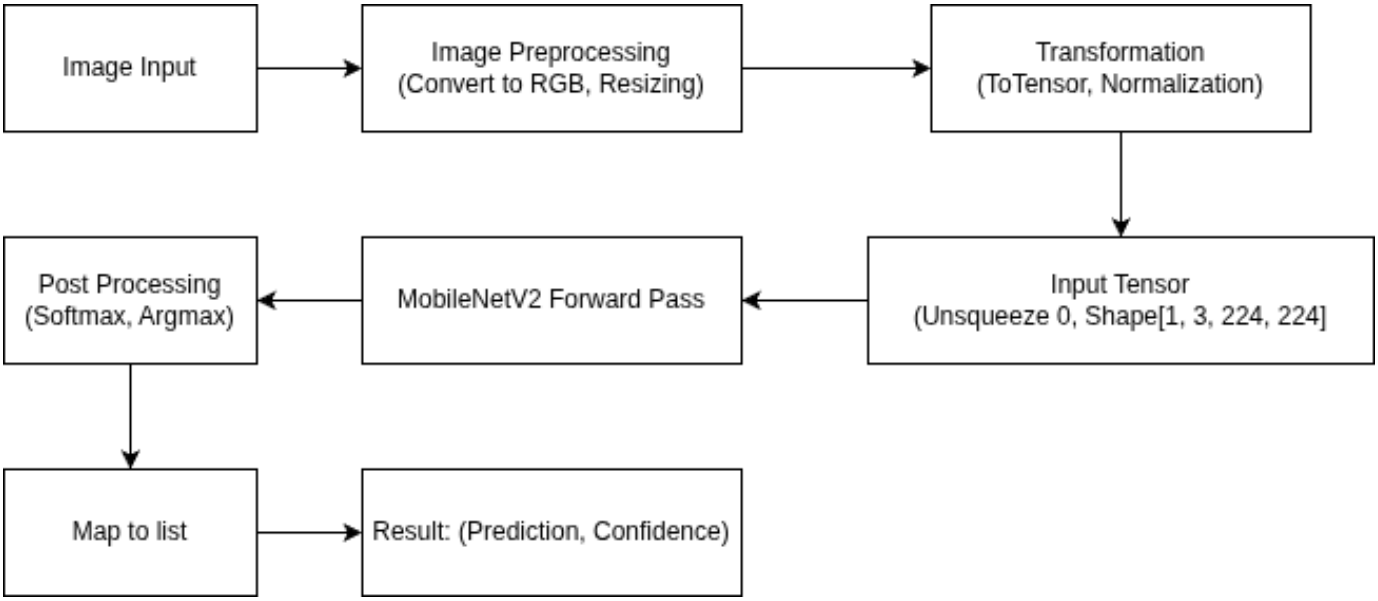


Fig. 3: Inference Pipeline of the Trained AGRIVISION Model

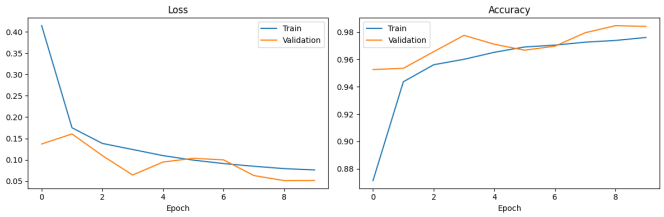


Fig. 4: Model Training Performance

B. Training and Validation

As illustrated in Fig 4, the model exhibits stable and consistent convergence throughout the training process. Both training and validation losses decrease steadily across epochs, while accuracy increases correspondingly. The close alignment between training and validation curves indicates that the model does not suffer from significant overfitting and is able to generalize effectively to unseen data.

TABLE I: Validation Accuracy Summary

Description	Accuracy (%)
Mean of Last 3 Epochs	98.28
Best Single Epoch	98.47

Table I summarizes the validation accuracy achieved by the model. The model records a mean validation accuracy of 98.28% over the final three epochs, with the best-performing epoch achieving 98.47% accuracy. These results confirm the stability of the training process and the robustness of the learned feature representations.

C. Classification Performance

Table II presents the overall classification performance of the proposed model across all 38 disease and healthy classes.

TABLE II: Overall Classification Performance

Metric	Value (%)
Accuracy	98.42
Precision (Weighted)	98.44
Recall (Weighted)	98.42
F1-score (Weighted)	98.41
Precision (Macro)	97.76
Recall (Macro)	97.74
F1-score (Macro)	97.73

The model achieves an overall accuracy of 98.42%, with weighted precision, recall, and F1-score all exceeding 98%. These high weighted metrics indicate strong performance across classes with varying sample sizes.

Macro-averaged metrics, which treat all classes equally, also remain high, with precision, recall, and F1-score exceeding 97%. This demonstrates that the model performs consistently well not only on majority classes but also on minority disease categories, highlighting its reliability for multi-class plant disease classification.

V. Discussion and Limitations

The experimental results demonstrate that the proposed MobileNetV2-based model achieves high classification accuracy and strong generalization performance on the PlantVillage dataset. The close alignment between training and validation metrics indicates that the model does not suffer from significant overfitting and is able to learn robust visual features for plant disease classification. Additionally, both weighted and macro-averaged precision, recall, and F1-scores consistently exceed 97%, confirming that the model performs well not only on majority classes but also across minority disease categories.

Analysis of the confusion matrix and per-class accuracy reveals that most classes achieve excelent performance,

particularly healthy leaves and diseases with distinctive visual symptoms. However, a small number of misclassifications occur between visually similar diseases on the same crop species, such as early blight and late blight in potatoes and tomatoes. These errors are expected due to overlapping visual characteristics, including similar lesion patterns and discoloration, which pose challenges even for human experts.

Despite its strong performance, the proposed system has several limitations. The PlantVillage dataset is collected under controlled environmental conditions with uniform backgrounds, which may not accurately reflect real-world agricultural settings. As a result, the model's performance may be affected when deployed in natural environments with varying lighting conditions, occlusions, or complex backgrounds. Additionally, the model performs single-image-based classification, which does not capture temporal disease progression and may limit diagnostic accuracy in early or ambiguous infection stages.

Overall, while the results validate the effectiveness of the proposed approach under controlled conditions, these limitations highlight important considerations regarding dataset diversity and real-world deployment constraints.

VI. Conclusion and Future Works

This study presents AGRIVISION, a plant disease detection system that leverages MobileNetV2 and computer vision techniques to support early and accurate identification of plant diseases. Using the PlantVillage dataset, the proposed model demonstrates high classification accuracy and strong generalization performance across 38 disease and healthy classes. The use of transfer learning enables efficient feature extraction while maintaining low computational cost, making the system suitable for deployment on web and mobile platforms.

By automating the disease detection process, AGRIVISION addresses key challenges associated with traditional manual inspection, including delayed diagnosis, human error, and limited accessibility for small-scale farmers. The system's image-based diagnosis, confidence scoring, disease history tracking, and community forum features collectively enhance usability and practical impact. Overall, this work highlights the potential of lightweight deep learning models to contribute meaningfully to sustainable agriculture and food security, in alignment with Sustainable Development Goal 2: Zero Hunger.

While the proposed system demonstrates strong performance under controlled conditions, several opportunities for improvement remain. Future work will focus on expanding the dataset to include real-world field images captured under diverse lighting conditions, backgrounds, and camera qualities to improve robustness in practical deployment scenarios. Additionally, incorporating temporal analysis using multiple images over time could enhance disease progression detection and early-stage diagnosis.

Further enhancements may include the integration of explainable AI techniques to provide visual interpretations of model predictions, thereby increasing user trust and transparency. The system can also be extended by refining the community forum with expert moderation and integrating localized agronomic recommendations based on detected diseases. Finally, continued optimization for edge devices and offline functionality will further improve accessibility for farmers in low-resource environments.

VII. Appendix

A. Team Contribution Statement

This project was completed collaboratively by a team of four members. Responsibilities were divided to ensure efficient development, thorough experimentation, and high-quality documentation. While each member had primary tasks, major design decisions and evaluations were discussed and agreed upon collectively.

1) Anthony Davidson Salim

- Contributed to dataset preparation, including image preprocessing, normalization, and data augmentation.
- Assisted in model training and validation analysis.
- Participated in system testing and performance evaluation.
- Contributed to writing the methodology and dataset description sections.

2) Kevin Nathaniel

- Led the design and implementation of the MobileNetV2-based model using transfer learning.
- Assisted in dataset preprocessing and hyperparameter tuning.
- Conducted performance evaluation and metric analysis.
- Contributed to the implementation and results sections of the report.

3) Rosaline Imely Gosang

- Developed and integrated the web-based user interface for image-based disease diagnosis.
- Assisted in model deployment on a Virtual Private Server (VPS).
- Contributed to backend integration and confidence score visualization.
- Participated in report review and system validation.

4) Joe Kenryoku Fendy

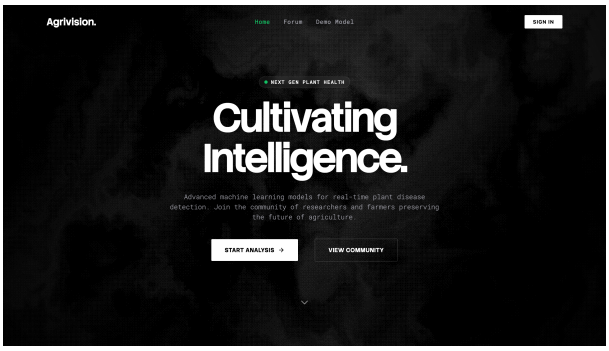
- Coordinated system integration and feature alignment across components.
- Assisted in data preprocessing and model testing.
- Managed documentation structure and report formatting.
- Contributed to the discussion, limitations, and conclusion sections

Shared Contributions

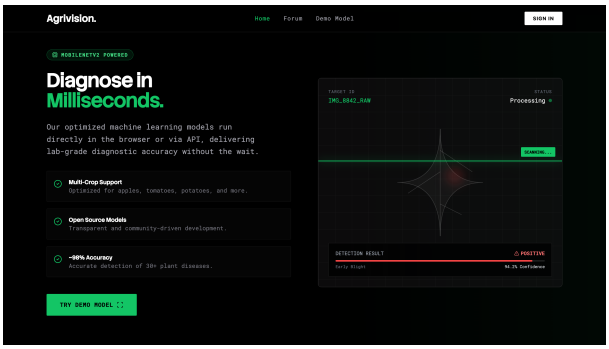
- Jointly defined project scope, system features, and evaluation strategy.
- Collaborated on debugging, testing, and result interpretation.
- Conducted peer reviews of code, experiments, and documentation.
- Finalized presentation materials and system demonstrations.

B. Screenshots

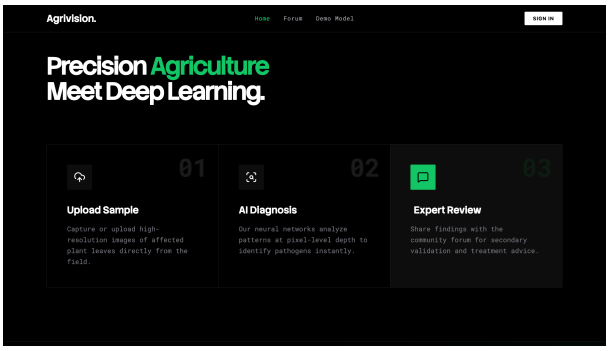
1) Home Page



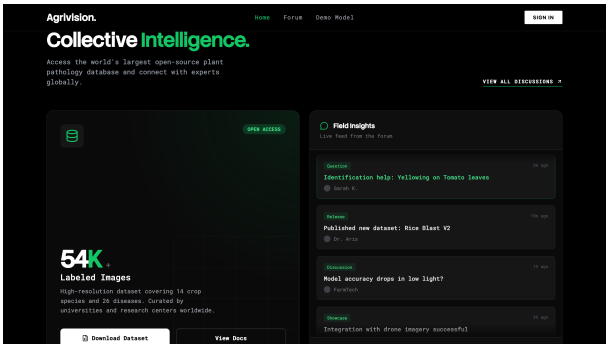
(a) Landing page hero section



(b) Real-time disease diagnosis visualization

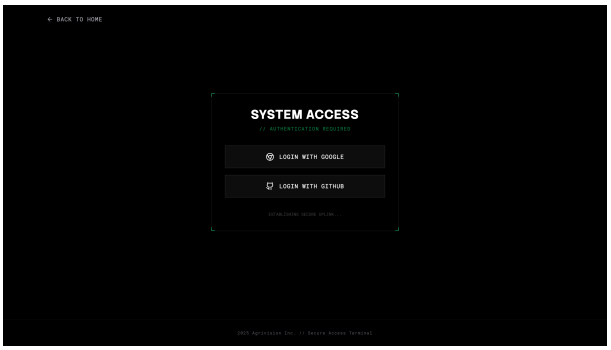


(c) Three-step user workflow for disease detection

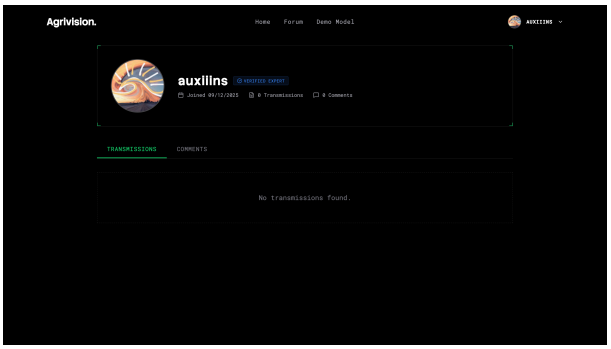


(d) Dataset statistics and community activity overview

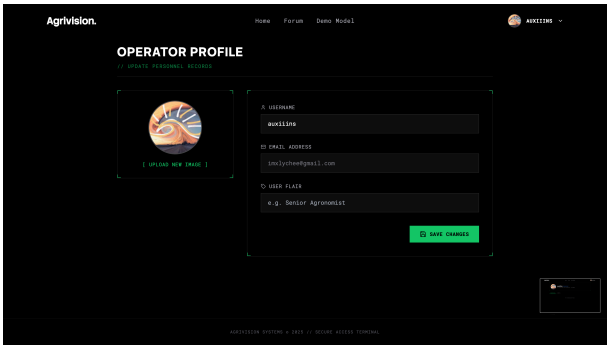
2) User Profile and Management



(a) Login page

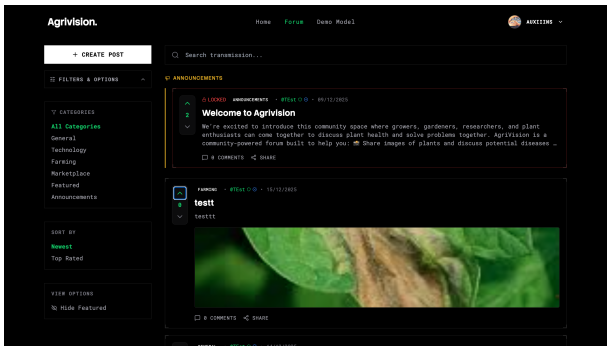


(b) User profile view

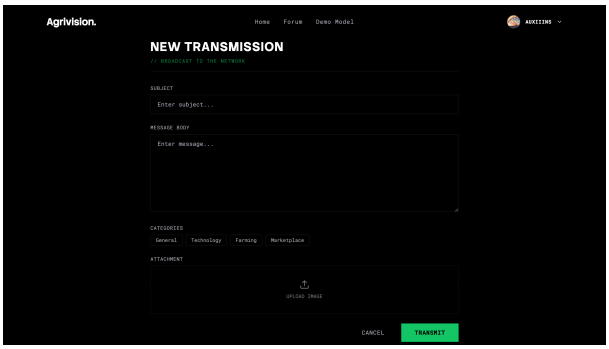


(c) Edit profile interface

3) Forum Page

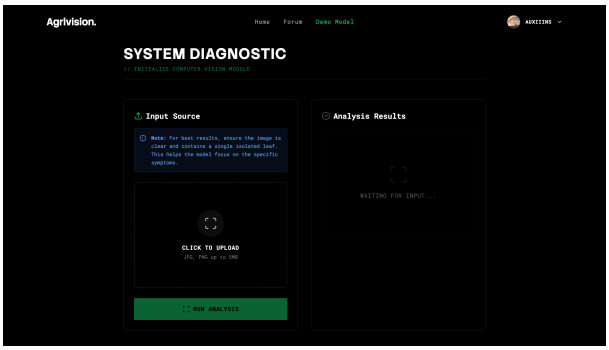


(a) Community forum feed showing user discussions

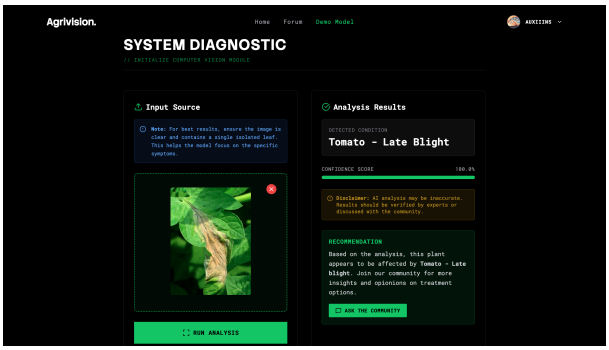


(b) Interface for creating a new forum post

4) Demo Model Page



(a) Image upload interface for plant disease detection



(b) Model inference results showing predicted disease and confidence score

The screenshots present the end-to-end functionality of the AGRIVISION platform, including the user workflow from image upload to disease prediction, secure user management, community-based discussion through forums, and real-time model inference with confidence scores. Together, these interfaces demonstrate the system's usability, practicality, and readiness for real-world deployment.

C. Code Snippets

You can access more of the code at https://github.com/KevinTheDuck/agrivision-cv-model.

```python
train_transform = transforms.Compose([
    transforms.Resize((CONFIG["image_size"], CONFIG["image_size"])),
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(10),
    transforms.ColorJitter(brightness=0.2, contrast=0.2),
    transforms.ToTensor(),
    transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225]),
])

val_transform = transforms.Compose([
    transforms.Resize((CONFIG["image_size"], CONFIG["image_size"])),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])
```

Fig. 5: Image Preprocessing

```python
model = models.mobilenet_v2(weights=models.MobileNet_V2_Weights.IMAGENET1K_V1)

for param in model.features.parameters():
    param.requires_grad = True

model.classifier = nn.Sequential(
    nn.Dropout(p=0.2),
    nn.Linear(1280, 128),
    nn.ReLU(),
    nn.Linear(128, CONFIG["num_classes"]),
)

model = model.to(DEVICE)
print(f"Model created with custom top layers:")
print(f"  - Global Average Pooling (1280 features)")
print(f"  - Dense: 128 neurons, ReLU activation")
print(f"  - Output: {CONFIG['num_classes']} neurons (Softmax)")
print(model.classifier)
```

Fig. 6: Model Definition

```python
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=CONFIG["learning_rate"])

print(f"Loss Function: CrossEntropyLoss (Categorical Cross-Entropy)")
print(f"Optimizer: Adam (lr={CONFIG['learning_rate']})")

def train_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    for images, labels in tqdm(loader, desc="Training"):
        images, labels = images.to(DEVICE), labels.to(DEVICE)

        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

    return running_loss / len(loader), correct / total
```

Fig. 7: Model Training

```python
def validate(model, loader, criterion):
    model.eval()
    running_loss = 0.0
    correct = 0
    total = 0

    with torch.no_grad():
        for images, labels in tqdm(loader, desc="Validating"):
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)

            running_loss += loss.item()
            _, predicted = torch.max(outputs, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()

    return running_loss / len(loader), correct / total
```

Fig. 8: Model Validation

D. Open Data

The dataset used in this study can be accessed through the following link: https://www.kaggle.com/datasets/mohitsingh1804/plantvillage

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